Predicting airport runway configurations for decision-support using supervised learning

Tejas G. Puranik
USRA, NASA Ames Research Center
Moffett Field, 94035, CA, USA
tejas.puranik@nasa.gov

Milad Memarzadeh
USRA, NASA Ames Research Center
Moffett Field, 94035, CA, USA
milad.memarzadeh@nasa.gov

Krishna M. Kalyanam
NASA Ames Research Center
Moffett Field, 94035, CA, USA
krishna.m.kalyanam@nasa.gov

***Abstract*—One of the most challenging tasks for air traffic controllers is runway configuration management (RCM). It deals with the optimal selection of runways on which to operate (for arrivals and departures) based on current and forecast traffic, surface wind speed, wind direction, other environmental variables, noise constraints, and several other airport-specific factors. In this paper, a methodology using supervised learning is developed to build a predictive model for RCM decision-support from large volumes of historical data. Data from two full years (2018 and 2019) related to current and forecast weather, demand/capacity, etc. is collected, analyzed, and fused together. A variety of supervised learning algorithms are tested for predicting runway configuration, and hyperparameter tuning is carried out to select the best performing model. The validation process involves two airports of low (Charlotte Douglas International Airport, CLT) and high (Denver International Airport, DEN) complexity of configuration decision-making. The results show significant promise for the two airports with test accuracy of 93% (CLT) and 73% (DEN). The methodology is scalable and generalizable to other airports across the U.S. National Airspace System.**

***Index Terms*—air traffic management, runway configuration management, machine learning**

I. Introduction

One of the key tenets of the next generation air transportation system (NextGen) is deploying automation capabilities through decision-support tools (DST) to aid air traffic controllers. Among such automation capabilities is the application of data-driven machine learning approaches to solve complex problems in air traffic management (ATM). One of the most challenging tasks for air traffic controllers is runway configuration management (RCM). It deals with the optimal selection of runways on which to operate (for arrivals and departures) based on traffic, surface wind speed, wind direction, other environmental variables, noise constraints, and several other airport-specific factors. RCM affects the efficiency of the National Airspace System (NAS) and both surface and airspace operations can benefit from a robust and accurate prediction of runway configurations.

There are two main types of analyses around RCM – *prescriptive* and *predictive* approaches. The main goal of prescriptive approach is to recommend the most optimal runway configuration for a given situation based on prevailing and/or forecast meteorological conditions, traffic load, intended system goals, and constraints. A sub-optimal selection of the runway configuration, or poor timing of configuration changes, can result in increase in taxi times for aircraft on the surface of the airport, excessive fuel burn due to holding patterns or increased taxi-times, and go-arounds for arriving traffic (e.g., due to excessive tail winds). The techniques in this type of application include heuristic search [1], [2], optimization and dynamic queuing theory [3]–[7], dynamic programming [8], and reinforcement learning [9]. Air traffic controllers and front-line managers are generally the intended stakeholders for prescriptive models.

Predictive models do not aim to improve existing decision-making rationale but rather try to learn it from historical data and mimic it with least amount of error. These models try to capture existing heuristics typically through supervised learning models [10]–[13]. Both air traffic management personnel and operators can benefit from predictive models through reduction of uncertainty in the prediction of airport capacity, better management of the traffic flow, and better utilization of resources. These two family of models are important and have received significant attention in literature, with more diverse applications seen in the prescriptive models [14].

Both prescriptive and predictive approaches generally involve building a model using historical operations data. This model is either based purely on the data (no site-specific adaptation required) or requires detailed airport-specific adaptation (such as a queuing network model [4], [5], [7]). Purely data-driven models have the advantage of not requiring airport-specific adaptation and tend to be extensible to other airports. On the other hand, a model that takes into account the dynamics of the airport movement can use simulation to optimize for the runway configuration selection or prediction.

While both predictive and prescriptive approaches have their own advantages and limitations as well as different applicability, this work focuses on predictive models. Recent literature in runway configuration prediction have used probabilistic approaches [3], [15] and/or supervised learning [12], [13] using limited number of variables such as the weather conditions, time of the day, and time of year. Other studies have developed a recursive multi-step forecasting model using supervised learning to predict the best choice of runway configuration given all the factors such as weather and future traffic [10]. Such methods use a vast amount of historical data to imitate the decision-making made by the controller

in the past with least amount of error. Other approaches have attempted to predict the arrival and departure runway assignments rather than airport configurations [11]. While these approaches present promise in the predictive setting, there is room for improvement in the prediction accuracy by incorporating other relevant features and/or developing more capable models.

There are several challenges associated with RCM predictive models due to which it is the subject of ongoing research. Air traffic controllers set the runway configuration based on relevant information and forecasting available at the time, including weather, traffic, noise abatement procedures, and safety bounds. This makes the process subjective due to potential biases in human decision-making as the same or similar conditions can lead to a different action by a different controller. There is also the challenge of inherent uncertainty in the accuracy of the available information, particularly weather forecast models and demand forecasts. An automated approach based on data mining can make use of the abundance of available historical data/decisions and learn through many more possible scenarios under uncertainty to provide the best estimate of runway configuration in the future.

In this paper, we present a comprehensive implementation of predictive models for runway configuration prediction from large volumes of historical data. Specifically, operational data from two full years (2018 and 2019) are collected, analyzed, and fused together to build the data product used in this work. The data set differs from prior work in the literature in terms of its scope, resolution, and variety of factors collected and considered. Meteorological data is collected from two different sources: current weather conditions from METAR (METeorological Aerodrome Reports) and forecast weather conditions from Localized Aviation Model Output Statistics (MOS) Program (LAMP). Operational data from the Federal Aviation Administration (FAA) Aviation System Performance Metrics (ASPM) related to scheduled and actual number of arrivals and departures, and average taxi time are collected. NASA's Sherlock Data Warehouse is used to identify other critical information such as go-arounds, and other events that might impact RCM decision-making. The overall goal is to learn, as closely as possible, decision-making in the historical data to predict for runway configuration for the future.

II. Methodology

Figure 1 shows an overview of the different elements of the methodology developed in this work. The remainder of this section details the elements of the methodology and how they are connected. One important objective of this work is to use relevant data from various NAS-wide sources and to train predictive RCM models that are purely data-driven, in that they do not require any airport-specific adaptation (e.g., geometry, procedural information, etc.). The methodology is setup in such a way as to make it repeatable and adaptable to other similar settings.

The methodology consists of three main parts which are expanded in this section – (1) Data Sources Identification and Extraction, (2) Data Processing and Transformation, and (3) ML Model Training, Validation, Testing.

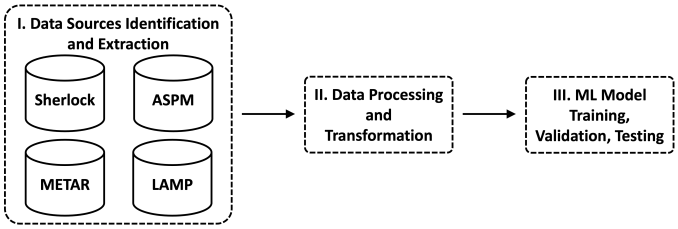


Fig. 1. High-level schematic of the methodology.

A. Data Sources Identification and Extraction

Four main data sources are used in this work, based on a survey of prior work. The data has been obtained from these sources and fused together using GMT time. The following subsections describe details of each data source utilized, the characteristics of the data, and the specific fields from the data that are used in this work. The four sources of data are:

(1) METAR (METeorological Aerodrome Reports)
(2) LAMP (Localized Aviation MOS Program)
(3) Sherlock Data Warehouse
(4) ASPM (Aviation System Performance Metrics).

Two types of meteorological data are used in this work which provide information about current weather conditions and forecast weather over the next several hours.

*1) **METAR**:* Prevailing weather conditions at an airport are available through the METeorological Aerodrome Reports[1]. Weather stations from around the world report weather conditions every hour using the METAR format. The data available in the database includes: temperature, dewpoint, wind speed, wind direction, visibility, clouds and ceiling, and any reported weather conditions.

*2) **LAMP**:* Weather forecast data is obtained from the LAMP system. LAMP is a statistical system that provides forecast guidance for sensible weather elements. LAMP updates MOS on an hourly basis, is run on National Oceanic and Atmospheric Administration (NOAA) Weather and Climate Operational Supercomputer Systems (WCOSS) computers and disseminated centrally, and provides guidance for over 1600 stations as well as gridded observation and forecast guidance on the NDFD continental U.S. 2.5-km grid out to 25 hours[2]. LAMP provides station guidance for the following weather elements: temperature, dewpoint, wind speed, direction, and gusts, probability of precipitation, precipitation type and characteristics, lightning, convection, ceiling height, opaque sky cover, visibility, and obstruction to

[1]https://www.aviationweather.gov/metar
[2]https://vlab.noaa.gov/web/mdl/lamp

vision.

*3) **Sherlock Data Warehouse**:* NASA's Sherlock Data Warehouse [3] contains information about flight events (such as go-around or missed approaches) that are useful for RCM prediction purposes. A go-around (or multiple go-arounds) could lead to a change of runway configuration and are therefore extracted from Sherlock and used in this work.

*4) **ASPM**:* The FAA's Aviation System Performance Metrics database[4] provides data on flights to and from the ASPM airports, and all flights by ASPM carriers, including flights by those carriers to international and domestic non-ASPM airports. All IFR and some VFR flights are included. ASPM reports also provide data on airport weather, runway configuration, and airport arrival and departure rates. While ASPM contains a large amount of detailed information and reports, in this work, the individual flight level data is downloaded for each airport. This contains detailed information on scheduled and actual times and delays for individual flights as well as the specific runway that the flight landed on. The information can then be aggregated up to the required time period (e.g. quarter hourly, hourly, etc.) to get the scheduled/actual counts of arrivals and departures and the average transit times during that time period.

*5) **Runway configuration (ASPM)**:* From the ASPM individual flight report, another important piece of information is derived: the actual runway configuration in use for each time period. This is obtained by first identifying the runways that are in use during the time period of interest based on the flights that landed and then aggregating them into standard configurations for use as labels in a machine learning model. The derivation of the different configurations is described here. It is noted that the runway configurations might be directly available from certain ASPM reports, however, in a small instance of cases, these reported configurations were found to be inaccurate for various reasons. Using the data of each of the individual flights that landed during a time period provides the highest level of fidelity around which runways or configurations were actually used.

There are two major configurations at Charlotte Douglas International Airport (CLT), the North-bound flow and the South-bound flow. In the North flow, runways 36L, 36C, and 36R are the primary runways for arrival and departure, while in the South flow, the opposite directions of these runways, i.e., runways 18R, 18C, and 18L, are the primary runways. A visual representation of the airport geometry is depicted in Figure 2 (left panel).

Denver International Airport (DEN) is an example of an airport with more complex runway configuration settings. As depicted in Figure 2 (right panel), it has six runways,

[3]https://sherlock.opendata.arc.nasa.gov/

[4]https://aspm.faa.gov/apm/sys/main.asp

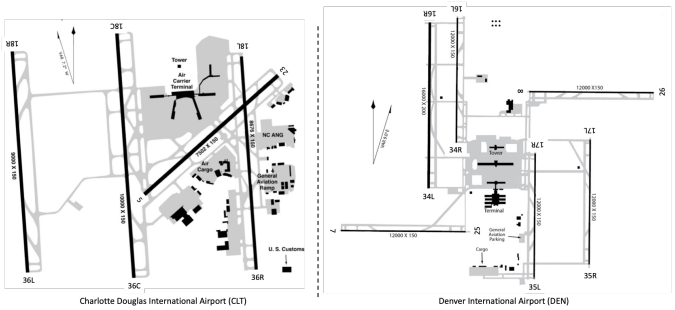


Fig. 2. Airport runway diagrams for CLT (left panel) and DEN (right panel).

among which four of them (34L/16R, 34R,16Lm 35L/17R, and 35R/17L) are North/South bound while the other two runways (7/25 and 8/26) are East/West bound. As a result, there are more combinations of runways that can be used for arrival/departure by the controller. Based on our comprehensive analysis of the data for years 2018 and 2019, and feedback from subject matter experts, eleven major configurations are identified, as reported in Table I. For example, for the configuration named N/NEW, North-bound runways (34R/L and 35R/L) are used for both arrival and departure, while runways 8 (East-bound) and 25 (West-bound) are used only for departure.

TABLE I
MAJOR RUNWAY CONFIGURATIONS FOR CLT AND DEN AIRPORTS.

Configuration [Arr/Dep]	Arrival Runways	Departure Runways	Usage Frequency [%]
CLT			
N/N	36C/L/R	36C/R	60.8
S/S	18C/L/R	18C/L	39.2
DEN			
SE/SE	16R/L, 17R/L, 7, 8	16R/L, 17R/L, 7, 8	18.8
S/S	16R/L, 17R/L	16R/L, 17R/L	15
N/NEW	34R/L, 35R/L	34R/L, 35R/L, 8, 25	14.5
S/SEW	16R/L, 17R/L	16R/L, 17R/L, 8, 25	12.6
N/N	34R/L, 35R/L	34R/L, 35R/L	12.3
NE/NE	34R/L, 35R/L, 7, 8	34R/L, 35R/L, 7, 8	11.7
NW/NW	34R/L, 35R/L, 25, 26	34R/L, 35R/L, 25, 26	8.6
SW/SW	16R/L, 17R/L, 25, 26	16R/L, 17R/L, 25, 26	3.4
E/E	7, 8	7, 8	1.6
NS/EW	34R/L, 35R/L, 16R/L, 17R/L	8, 25	1.2
W/W	25, 26	25, 26	0.3

B. Data Processing and Transformation

From among the data sources identified, there are several parameters available for use in the machine learning models.

The first step is to identify and retain any parameter that might impact runway configuration decision-making and discard the other parameters. This selection is a conservative filtering and many of the parameters are retained. The next step is to identify and mitigate data quality issues, perform data transformations, and convert text data into numeric values. Data transformation is an important step as it translates the raw data into features and labels for use in a machine learning model. Data processing involves several key steps such as aggregation of data to the frequency being considered (15-minutes intervals in this work), filling missing data, and encoding categorical variables. Variables deemed important, but that are unavailable or missing in most of the data need to be excluded. Data processing and transformation thus convert the raw data from various sources into features that can be used for machine learning. Through the data processing and transformation step, each 15-minute interval in the data available is associated with a set of features and a unique runway configuration. The following two subsections identify the features used in the final model (including the rationale) and the labels for supervised learning.

1) Features: Table II shows various features used in the machine learning (ML) model for predicting runway configuration. The table contains the feature name, the data source used for building this feature, and the feature type (categorical, continuous). The first set of features are derived from the METAR data. The wind speed is a continuous feature and wind direction is discretized into eight distinct categories, which are then encoded using one-hot encoding. Wind speed and direction are important variables that affect the airport configuration as high tailwinds and crosswinds are operationally unsafe and can render certain configurations unusable. Temperature, ceiling, and visibility are other weather features that are included from the METAR data and are continuous variables. The meteorological condition is a binary variable having two possible states: Visual Meteorological Conditions (VMC) and Instrument Meteorological Conditions (IMC). It indicates the prevailing conditions based on the combination of visibility and the cloud ceiling.

The next set of features are those derived from the ASPM data. This includes the scheduled arrivals and departures, which give a sense of the demand at the airport. Demand is an important factor that affects possible airport configurations as high-demand situations would require high-capacity configurations to be used. The capacity of arrivals and departures is the next feature used from ASPM and it provides information about the number of arrivals that a configuration can accommodate. Capacity is also typically correlated with various weather conditions (particularly VMC versus IMC). ASPM also contains the count of actual arrivals and departures, however, those are not used in this work. Similarly, averaged taxi-in and taxi-out times are also available from ASPM but these are not used because they are an aggregated metric that would typically not be available in real time during operations. The scheduled arrivals and departures over a 1-hour window following the current time interval are generated and used as features rather than the scheduled operations in the current time interval. The local hour and quarter hour is also available and is encoded as one-hot categorical variables in order to account for variations during the day.

TABLE II
VARIABLES USED IN FEATURES IN THE FUSED DATA PRODUCT USED FOR MACHINE LEARNING MODEL.

Feature	Type	Source
Wind Speed	Continuous	METAR
Wind Direction	Categorical	METAR
Ceiling, Visibility, Temperature	Continuous	METAR
Meteorological condition	Binary	METAR
Scheduled arrivals and departures	Continuous	ASPM
Capacity of arrivals and departures	Continuous	ASPM
Local hour	Categorical	ASPM
Quarter hour	Categorical	ASPM
Number of go-arounds	Continuous	Sherlock
Multiple go-arounds	Binary	Sherlock
Wind Speed Forecast	Continuous	LAMP
Wind Direction Forecast	Categorical	LAMP
Forecasts of Ceiling, Visibility, Temperature	Continuous	LAMP
Precipitation Forecast	Categorical	LAMP
Convection Forecast	Categorical	LAMP
Lightning Forecast	Categorical	LAMP
Meteorological Condition Forecast	Binary	LAMP

Two features are created from event data extracted from the Sherlock data warehouse: the total number of go-arounds in a time period and the presence of multiple go-arounds in a time period. This could be an important feature, because high tailwinds or crosswinds could be one of the reasons for a go-around and one or more go-arounds could prompt the controller to change the runway configuration earlier than planned.

The final set of features is obtained from the LAMP meteorological forecast data. LAMP features contain forecasts for wind speed, wind direction, ceiling, visibility, temperature, and meteorological condition. Additionally, LAMP also provides forecasts of precipitation, convective weather, and lightning as categorical variables whose values range from 0 to 4, with increasing numerical value indicating higher probability. Forecast weather elements are an important part of predictive runway configuration models, as they can provide insight into future weather changes and potential for switches in configurations. Inclement weather forecast can lead to reduced capacity and changes in schedule as well as the exclusion of certain runway configurations. The LAMP data features help account for these scenarios and other implicit decision-making strategies that may be employed by controllers when they review the forecast. Since the LAMP data is available at an hourly interval, the data product in this work will utilize the most recently available forecast for each 15-minute interval. While the forecast data is available up to 25 hours in the future, only the next 2 hours are currently utilized in the ML

model, as these are seen as the most relevant to the prediction of configurations in the immediate next 15-minutes interval.

As is evident from the description of the features as well as their impact on the runway configuration management problem, there exist correlations and coupling between different features used in the ML models. Dimensionality reduction techniques such as Principal Component Analysis (PCA) are explored in the implementation section to improve performance.

2) Labels/Targets: The labels or targets for the machine learning model are the unique runway configurations for each airport. Thus, CLT has 2 unique labels (N or S) and DEN has 11 unique labels (identified in Table I). The runway configuration in the next 15-minutes time interval is the label associated with the current interval. In this way, during operations, the model is using features available at the current time to make the prediction for the future without any leakage of data.

C. ML Model Training, Validation, Testing

The end product generated from data processing and transformation is used to setup and train the machine learning models. The overall setup of the model is shown in Figure 3. At any point in time, the model uses features from the current step to predict the runway configuration for the next step (next 15-minutes interval). Once trained, the model runs continuously to provide predictions for the next time step (dashed box and arrows). It is noted here that features in the current time step might include future estimates of some quantities (such as weather forecast from LAMP), but these data are available or collected at the current time step or earlier.

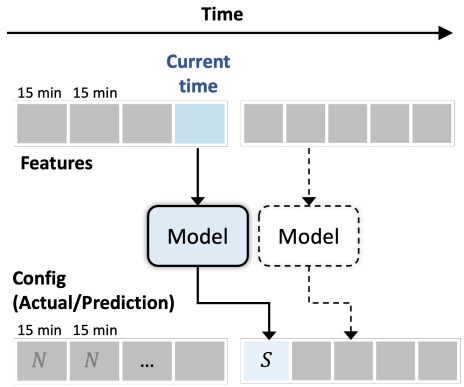


Fig. 3. High-level schematic of the methodology.

Several different machine learning classification models were considered as candidate models for this work including Random Forest, Gradient Boosting, Support Vector Machine, etc. Among these models considered, Random Forest had the best overall performance at both airports and therefore, in this paper, results from the best performing (after tuning the hyperparameters) Random Forest model are shown. Standard machine learning practice is used for training, validation, and testing of the models. The overall data set is randomly divided into two sets for training (80%) and testing (20%). The model hyperparameters are tuned using an off-the-shelf python package called *GridSearchCV*[5] that finds the optimal hyperparameters and performs 5-fold cross validation based on the training set. For the Random Forest algorithm, the following hyperparameters are varied based on review of literature [16] and prior usage of this algorithm – the number of estimators, maximum depth, minimum number of samples per leaf. For details on each of the hyperparameters and their impact on the model, readers are referred to the documentation of Random Forest model implementation used in this work (from the python library scikit-learn[6])

III. Results and Discussion

In this section, the described methodology is implemented and the results are reported for the two airports of interest (CLT and DEN).

A. Data Description and Exploration

The data used in this work is collected from the data sources described earlier for two full years (2018 and 2019) at CLT and DEN airports. The data product is created using data processing and transformation to produce one row for each 15-minutes interval during the two years. The sources that are available at a lower frequency are up-sampled by using data from the closest past instance. Thus, each row in the final data product is one instance of the 15-minutes interval and the columns are the features identified from previous sections. The final data product is stored and used as a *DataFrame* object in the pandas library in python[7]. The DataFrame for each of the two airports consists of 70,056 rows. For each airport, 69,480 rows ($\approx$ 99.1%) do not contain any missing data or features and are thus used in the model training, validation, and testing. The total number of columns available from the raw data set is 212, however, only a subset of these columns are used as features in the ML model. A total of **84** features are used in the final ML models in this work. These **84** features correspond to **30** variables derived from Table II (the higher number of features than variables is due to the encoding of categorical features that leads to multiple features for each variable). The data is then divided into training (80%) and testing (20%) sets based on random splits. In each of the following sections, the scores or metrics are all reported on the test set unless explicitly stated otherwise.

B. Optimal ML models

Random Forest with grid search is then implemented for each data set (CLT and DEN) to obtain the best performing models. Typically, grid search will select the best hyperparameter set based on a scoring criteria such as accuracy, precision,

[5]https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.GridSearchCV.html

[6]https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html.

[7]https://pandas.pydata.org/

and/or recall. In cases where the data is balanced (approximately similar number instances of each class), accuracy is a sufficient score. However, in cases such as the current work, the data labels are unbalanced. For example, some configurations might be used more than others at an airport (see Table I). Therefore, in this work, the F-1 score (harmonic mean of precision and recall) is used as the scoring metric in grid search as it explicitly accounts for class imbalance. Five-fold cross validation is used in the hyperparameter tuning, and the folds are made by preserving the percentage of samples for each class (called stratified folds). Dimensionality reduction through the PCA method was tested but did not yield any significant performance improvements.

1) Hyperparameters: The hyperparameters for the best performing models after tuning and their ranges are reported in Table III for both airports. A short description of the parameters is provided here, and readers are referred to scikit-learn[7] documentation. The number of estimators is among the main hyperparameters in the model, and it controls the number of decision trees in the forest. The next hyperparameter is maximum depth of a decision tree. It controls how far down the nodes are expanded for each tree. The third hyperparameter is the max samples. It controls the maximum number of samples to draw to train each decision tree. Values lower than one (the max value for this hyperparameter) ensure that the different decision trees trained have different strengths that can help with model performance. All three of these hyperparameters help to achieve the desired trade-off between model performance and generalizability. It is noted that the optimal values of the hyperparameters can vary when the data is changed, even for the same airport. The values provided in the table gave the best performance for the available data.

TABLE III
OPTIMAL HYPERPARAMETER VALUES FOR RANDOM FOREST MODELS.

Hyperparameter	CLT Value	DEN Value	Min	Max
Number of estimators	300	400	50	500
Max. depth	25	50	25	200
Max. samples	0.7	0.8	0.7	1.0

2) Model Performance Metrics: Table IV shows the overall performance of the best models from Table III for each airport. The training and test accuracy are provided along with the F-1 score for each model. For multi-label classification (such as the case with DEN airport here), the F-1 score is obtained for each label individually and is generally averaged to obtain a single score for the model. The averaging can be done in various ways; in this work, *weighted* averaging is used. For *weighted* averaging, the metrics needed to calculate F-1 score (precision, recall) are calculated for each label, and their average is computed weighted by the number of true instances for each label.

As evident from the table, the optimal ML model performs well for both airports, with CLT performing better than DEN. This is an expected outcome, as the level of operational complexity at CLT is lower and the ML model only has to predict from among *two* configurations as opposed to *eleven* for DEN. The metrics obtained using the trained ML models are also compared against a baseline that is based on air traffic controllers' past decisions. This baseline is obtained by selecting the most common action taken by the controllers in the past for each similar state (unique combination of wind direction, wind speed, hour of the day, meteorological condition, etc.). The average accuracy for this baseline model that mimics the most frequent controller action is 81.6% for CLT and 53.7% for DEN, which shows the considerable variations in the controllers' decision-making. If the controllers always made the same decision in the same specific state in the historical data, these numbers would be 100%.

TABLE IV
OVERALL MODEL PERFORMANCE.

Airport	Training Accuracy %	Test Accuracy %	Test F-1 Score (weighted) %
CLT	98.94	93.94	93.96
DEN	96.45	73.49	74.05

3) Confusion Matrix and Feature Importance: Figures 4 and 5 show the confusion matrix for the performance of the ML model at CLT and DEN airports. The actual class/configuration represents the historical decisions made by the controllers and is shown on the y-axis, while the predicted class/configuration by the ML model is shown on the x-axis. Diagonal elements of the matrix show correct predictions, while the off-diagonal elements depict errors made by the ML model. The color of each box in the matrix represents the amount of samples that belong in that box with darker color indicating more samples.

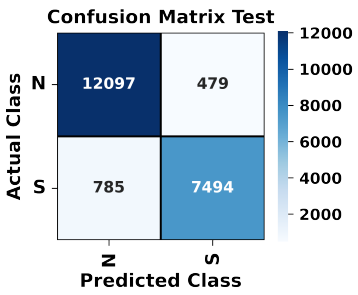


Fig. 4. Confusion matrix for CLT model.

For CLT, the model does a good job of identifying the right configuration in a majority of cases. Because North flow is preferred in the operations and is thus better represented in the data, it has a better performance in the model. In the case of DEN, it can be seen that a majority of the disagreements are among very similar configurations. For example, in the case of S/S configuration, majority of the times that the ML model disagrees with the historical decisions is in prediction of SE/SE instead of S/S, which are very similar configurations. Another example is the confusion of the algorithm between NE/NE and N/NEW configurations. Some of these disagreements are

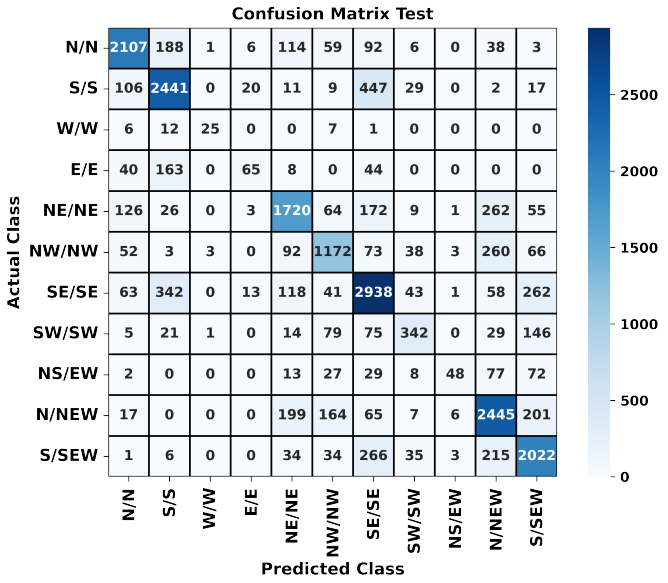


Fig. 5. Confusion matrix for DEN model.

due to the presence of class imbalance in the data as noted in Table I and can be improved by additional data collection and training of the model with a larger amount of data. Moreover, the use of class balancing techniques such as over-sampling and/or under-sampling can potentially improve performance of the model, however, it is out of the scope of this paper and will be investigated in the future. On the other hand, some of the disagreements happen because factors other than those represented in the feature space might be playing a role in the decision-making around configurations. This might lead to two states in the feature space that are very similar to each other to have different actual configuration in the historical data.

Additionally, for ensuring appropriate behavior in edge-cases, a metric for violation of obvious decisions is defined. For this metric, decisions that are obvious to the subject matter experts are defined as cases. Then, the percentage of times the model violates these defined cases is identified. For example, in the case study of CLT, if the wind is blowing strongly (more than 15 knots) from North, if the model suggests the South configuration, it is considered a violation. For both models, very low percentages of violation metric (below 0.07% and 0.7% for CLT and DEN) are observed.

Figures 6 and 7 provide an overview of the most important features as perceived by the Random Forest algorithm. This feature importance is calculated by a method known as mean decrease in impurity. In simple terms, it counts the times a feature is used to split a node, weighted by the number of samples it splits. One of the drawbacks of this method is that the importances can be high even for features that are not predictive of the target variable, as long as the model has the capacity to use them to overfit[8]. Therefore, these importances should be examined from the perspective of understanding the algorithm's high-level behavior rather than drawing any detailed inferences from them.

[8]https://scikit-learn.org/stable/auto_examples/inspection/plot_permutation_importance.html

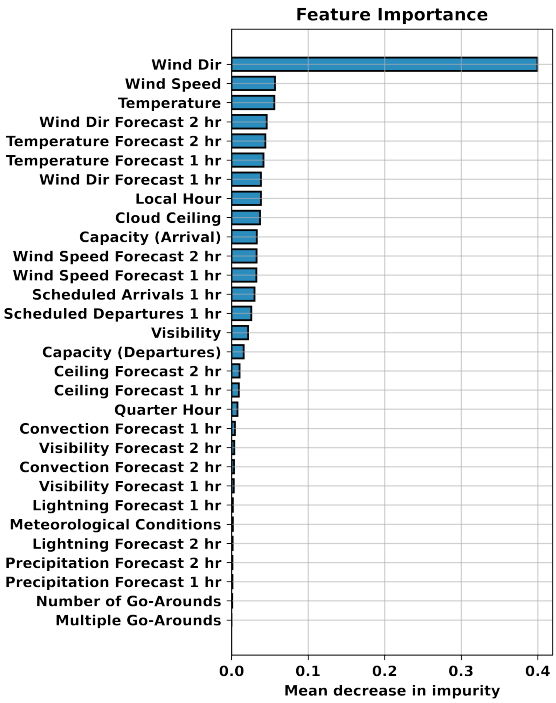


Fig. 6. Feature importance for CLT model.

For CLT airport, current wind direction dominates the feature importance chart followed closely by the current wind speed indicating that the North/South configuration choice is heavily influenced by wind. Among other important factors are forecasts of wind direction over the next 1 and 2 hours. Current and forecast temperature is an unexpected parameter among the top ones with no clear explanation and is likely an artifact of the way the feature importance is calculated rather than any insightful reason.

For DEN airport, the feature importances are more spread out over the top few factors as opposed to a single dominating factor in CLT. The scheduled arrivals and departures in the next hour (demand) and the wind direction are among the top factors. The local hour is also among the top three indicating that there are perhaps preferred configurations for specific times of the day. Wind speed and other forecast variables are also of higher importance than in CLT.

Forecasts of other variables, meteorological conditions, and go-arounds seem to surprisingly play little role in the ML model for both airports.

C. Discussion

It is observed that the Random Forest model generally performs well for both CLT and DEN airports. Based on the metrics and feature importances, it is evident that the performance of the model depends on the airport complexity and operational constraints at each airport. The benefits of

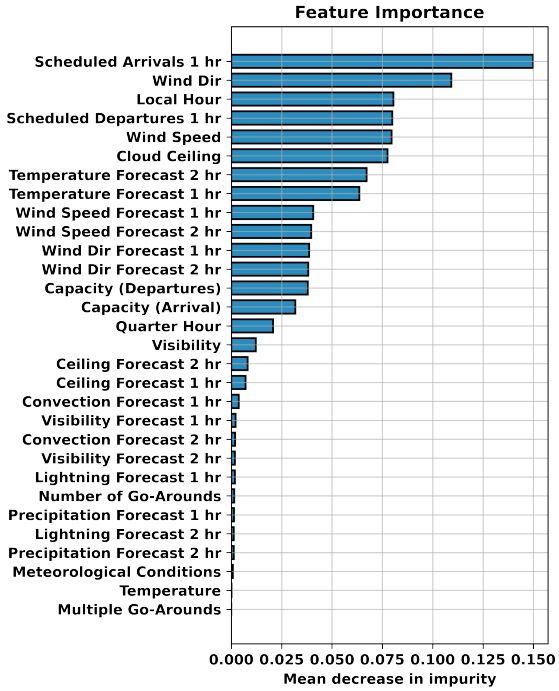


Fig. 7. Feature importance for DEN model.

using a large number of features is evident from the two feature importance figures where the most important features are quite different across two airports. As the methodology is expanded to other airports with more diverse operating conditions, other features might play an elevated role. The data sources used in this work are easily available for a majority of large airports across the NAS and therefore the methodology lends itself easily to expansion at other airports. The extraction of labels (actual runway configurations in use) is an important and time-consuming step that was essential to ensure most accurate information. Performing this at each airport might require some amount of initial offline work. Possible alternative data sources or streams might need to be explored to scale this methodology. Additionally, the drop in performance of the model without performing this step can be explored to gauge the quantitative benefits of doing this step. This will enable making a decision on whether there is added value in performing this time-consuming offline analysis. Hyperparameter tuning, while important only marginally improves a baseline model built with default hyperparameters from scikit-learn. This indicates that the model is performing as well as it can with the features that have been provided to it. Further improvement might require a different model type or additional features that cover other aspects of operations. Finally, it was observed that there is a variability in the historical decisions by controllers when the system is in the exact same state (weather, forecast, traffic, etc.). This variability can be captured using a measure known as entropy from information theory. Entropy allows quantifying how much variability exists in the controllers' decisions in any given state and thereby provides some context on the supervised learning model's predictions.

IV. Conclusion

In this paper, a methodology for predicting airport runway configurations using machine learning algorithms trained on data collected from various sources was demonstrated. A detailed description of data sources and processing required for the methodology was provided along with a description for obtaining labels for the ML model. Various ML models were explored and the best performing algorithms and hyperparameters were reported for two representative airports of low and high operational complexity (CLT and DEN respectively).

There are some identified avenues of future work that would improve the methodology further. The first avenue identified is the inclusion of other weather forecast data called Terminal Aerodrome Forecast (TAF). This is a popular aviation weather data that is frequently used by air traffic personnel in the decision-making process. The comparison of using LAMP versus TAF data will be explored. The second avenue is expansion of the methodology to other airports. This would enable learning how the methodology scales and identify gaps in the current data or model setup, which might help models at existing airports as well as new ones. The third avenue of future work is exploring more complex machine learning models such as deep learning or long short-term memory (LSTM) networks that can explicitly account for the temporal nature of the data and potentially help realize additional gains in the model performance.

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